# A MAJOR PROJECT REPORT

## ON

# ENHANCEMENT OF VIRTUAL ASSISTANT THROUH MULTIMODAL AI FOR EMOTION RECOGNITION

**Submitted to**

**SRI INDU COLLEGE OF ENGINEERING AND TECHNOLOGY**

In partial fulfillment of the requirements for the award of degree of

**BACHELOR OF TECHNOLOGY**

**In**

**ARTIFICIAL INTELLIGENCE AND DATA SCIENCE**

**Submitted**

**By**


**D. KAVITHA SRI - (21D41A7214)**

**B. SHARATH - (21D41A7204)**

**B. AJAY - (21D41A7203)**

Under the esteemed guidance of

**Mr. R. MADHAVA REDDY**

(Assistant Professor)

**DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE**

**SRI INDU COLLEGE OF ENGINEERING AND TECHNOLOGY**

(An Autonomous Institution under UGC, accredited by NBA, Affiliated to JNTUH)

Sheriguda,Ibrahimpatnam.


(2024-2025)

# SRI INDU COLLEGE OF ENGINEERING AND TECHNOLOGY

**(An Autonomous Institution under UGC, Accredited by NBA, Affiliated to JNTUH)**

**DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE**

## CERTIFICATE

Certified that the Major Project entitled "**ACCESS CONTROL BY SIGNATURE KEYS TO PROVIDE PRIVACY FOR CLOUD AND BIG DATA**" is a Bonafide work carried out **by D. KAVITHA SRI(21D41A7214), B. SHARATH(21D41A7204), B. AJAY (21D41A7203)** in partial fulfillment for the award of **Bachelor of Technology** in **ARTIFICIAL INTELLIGENCE AND DATA SCIENCE** of **SICET**, Hyderabad for the academic year 2024- 2025. The Project has been approved as it satisfies academic requirements in respect of the work prescribed for **IV YEAR, II-SEMESTER** of **B. TECH** course.



| | |
|---|---|
| **INTERNAL GUIDE** | **HEAD OF THE DEPARTMENT** |
| **(Mr. R. Madhava Reddy)** | **(Dr. T. Charan Singh)** |
| **(Assistant Professor)** | **(Associate Professor)** |

**EXTERNAL EXAMINER**

# ACKNOWLEDGEMENT

With great pleasure we want to take this opportunity to express our heartfelt gratitude to all the people who helped in making this project work a success. We thank the almighty forgiving us the courage & perseverance in completing the project.

We are thankful to principal **Prof. Dr. G. Suresh**, for giving us the permission to carry out this project and for providing necessary infrastructure and labs.

We are highly indebted to, **Dr. T. Charan Singh**, Head of the Department of Artificial Intelligence and Data Science, Project Guide, for providing valuable guidance at every stage of this project.

We are grateful to our internal project guide, Mr. **R. MADHAVA REDDY** for his constant motivation and guidance given by his during the execution of this project work.

We would like to thank the Teaching & Non-Teaching staff of Department of Artificial Intelligence and Data Science for sharing their knowledge with us.

Last but not the least we express our sincere thanks to everyone who helped directly or indirectly for the completion of this project.

| | |
|---|---|
| **D. KAVITHA SRI** | **- (21D41A7214)** |
| **B. SHARATH** | **- (21D41A7204)** |
| **B. AJAY** | **- (21D41A7203)** |

# INDEX

# 1. INTRODUCTION

Cloud computing and Big data as novel techniques need more attention and research. The privacy for these novel techniques is one of the most important issues. Shared data or processing and transferring data in third party could is more vulnerable to attacks, such as sync cookies, attacks on client profiles, limited connections of provided, etc. [1]. Cloud computing is seen as an essential, low-maintenance way to share resources. More and more, moving the systems that manage the local information into cloud servers has become the standard procedure, clients being able to benefit of premium services whilst saving big money on the local infrastructures. Users that use cloud computing are no longer faced with the disadvantages of the problematic local solutions for storing and management of data. Using policy of encrypted data based on access control is the most common privacy method. This policy cab ensures privacy of data that represent sensitive information.

The privacy based on access control means to allow access to data only to authorized persons. The access mechanisms to the sensitive data have problems if they can be shared without strong privacy. Data is often in cloud or big data with shared access with third party, which makes it more vulnerable to attacks. Usually, moving data between sides can be risky on client privacy. To ensure end-to-end security, we try to implement algorithms to provide strong privacy for big data in cloud. Other related works and research in same area, as the encryption based-attribute, show the most suitable approach to identify efficient and more scalable methods. Cloud computing implies a set of computers that are used together to provide different accounts and services. The benefits of using cloud computing in companies are cost reduction and time saving. Also, using shared services from cloud is easier than building and developing own infrastructure. The providers of cloud computing are focus on providing flexible services, cost-effective IT infrastructure and secure environments for companies and organizations [4].

The main issue with big data in cloud is that processing or usage always needs to be done by third party. It is very important for the owners of data, or clients, to trust and to have the guarantee of privacy for the information stored in cloud or analyzed as big data. The privacy methods studied in previous research showed that privacy infringement for cloud computing and big data happened because of limitation, privacy guarantee rate or risks on data by external or internal attackers. Generally, the private client data has been under attack.

# 2. LITERATURE SURVEY

**TITLE:** A Privacy Preserving System for Cloud Computing.

**AUTHOR:** Ulrich xzzzq, Benjamin Justus, Dennis Loehr,(2011),

**ABSTRACT:** Cloud computing is changing the way that organizations manage their data, due to its robustness, low cost and ubiquitous nature. Privacy concerns arise whenever sensitive data is outsourced to the cloud. This paper introduces a cloud database storage architecture that prevents the local administrator as well as the cloud administrator to learn about the outsourced database content. Moreover, machine readable rights expressions are used in order to limit users of the database to a need-to-know basis. These limitations are not changeable by administrators after the database related application is launched, since a new role of rights editors is defined once an application is launched. Furthermore, trusted computing is applied to bind cryptographic key information to trusted states. By limiting the necessary trust in both corporate as well as external administrators and service providers, we counteract the often criticized privacy and confidentiality risks of corporate cloud computing.



**TITLE:** Enhance Big Data Security in Cloud Using Access Control.

**AUTHOR:** Yong Wang, Ping Zhang (2017),

**ABSTRACT:** Big data has become an essential requirement for enterprises looking to harness their business potential. The use cases for big data are endless and range from customer targeting and fraud analytics to anomaly detection and more. This data can be generated quickly from various sources such as users' browser and search history, credit card payments, mobile pinging of the nearest cell phone tower, etc. Given the volume of sensitive information being captured, any or accidental disclosure of or access to the data can have severe consequences for your enterprise, both in financial terms and in more intangible ways, such as the loss of brand recognition and users' trust.

In recent years, many highly scalable and complex processing *frameworks for big data* have emerged, such as Hadoop, Hive, Presto, and Spark. Securing these frameworks is very challenging because of their distributed nature, and involves many touchpoints, services, and operational processes. With accelerating cloud adoption, monitoring access and data flows for big data becomes even more complicated.

**TITLE:** Security Techniques for Data Protection in Cloud Computing.

**AUTHOR**: Kire Jakimoski, (2016),

**ABSTRACT:** The IBM Security® Guardium® Insights data security platform allows any enterprise to quickly address their data security and compliance needs. Its robust capabilities help enterprises automate compliance policy enforcement and centralize data activity from multiple clouds. This enhances visibility to achieve a consolidated view of how critical data is being accessed and used across hybrid environments. Guardium Insights is designed to provide data security specialists with a centralized hub where they can retain monitoring data for compliance or analytic purposes for as long as they need to. With best- in-class features such as automated compliance, audit and reporting, and real-time monitoring, Guardium Insights can help users achieve a reduction in audit prep time. Guardium Insights can complement and enhance existing Guardium® Data Protection deployments or be installed on its own to help solve compliance and cloud data activity monitoring challenges.



**TITLE**: Toward cloud-based collaboration services

**AUTHOR:** Banks, David, John S. Erickson, and Michael Rhodes. (2009)

**ABSTRACT:** As we approach the end of the first decade of the 21st century, we are witnessing a disruptive change in the provisioning of information technology: the advent of cloud computing. For most organizations, information technology is not a core competence. Until recently, their only option was to retain IT specialists on premise, but now alternatives from the likes of Google, Amazon and Salesforce.com are becoming increasingly viable. Accordingly, out-sourced IT is now an option for all sizes of company. This coincides with businesses seeking to operate efficiently in a global marketplace by outsourcing noncore competencies. As businesses choose to excel in a single area and partner for the rest, collaboration across organizational boundaries becomes a core part of product development. Traditional Enterprise Content Management (ECM) software has not kept up, leaving people collaborating via email—the lowest common denominator. In response to these trends, we envision a generation of cloud-based collaboration platforms emerging to address the needs of content-centered collaboration between businesses. millions of users. The Fractal research program [1] in HP Labs aims to design and deploy such a platform.

# 3. SYSTEM ANALYSIS

## 3.1 EXISTING SYSTEM

Cloud computing is seen as an essential, low-maintenance way to share resources. More and more, moving the systems that manage the local information into cloud servers has become the standard procedure, clients being able to benefit of premium services whilst saving big money on the local infrastructures. Users that use cloud computing are no longer faced with the disadvantages of the problematic local solutions for storing and management of data. Using policy of encrypted data based on access control is the most common privacy method. This policy cab ensures privacy of data that represent sensitive information. The privacy based on access control means to allow access to data only to authorized persons. The access mechanisms to the sensitive data have problems if they can be shared without strong privacy.

## DISADVANTAGES

1. Data is often in cloud or big data with shared access with third party, which makes it more vulnerable to attacks. Usually, moving data between sides can be risky on client privacy.
   **Key Distribution:** Distributing signature keys securely to authorized users or entities can be challenging. If not handled properly, it can lead to key exposure or unauthorized access. Implementing secure key distribution mechanisms is crucial
   **Human Error:** Users may accidentally mishandle their signature keys, leading to access problems or security vulnerabilities. Training and user education are essential to mitigate this.

## 3.2 PROPOSED SYSTEM

In our paper we are forwarding a case study that is built on the basis of three models: first model consists of the cloud's architecture, which will contain all the transactions for the other models; the second model is based on the concept of transactions' manager, who provides Keys, grand users, manages the queries and so forth; and finally, the last model is the one concerned with the clients, i.e. the staff that already has the right to use data in the cloud or analysis of big data. The cloud architecture is managed by providers of services. In addition, we consider that the case study is based on the assumption of zero trust between the three models. This situation is due to the fact that all transactions will be carried out by a third party and data movements through various levels of security. Among the tasks of the transactions' manager we might enumerate: user registration, generation of system parameters, user revocation, and the verification of the identity of data owners. The clients' model is a dynamic one, depending on the kind of transactions and data used.

## ADVANTAGE OF PROPOSED SYSTEM

- They proposed protecting from concerns of attacks on the data stored in the cloud using model segregation of the data.
- The data value in cloud gained during acquisition is separated in multi-location to support privacy of clients.
- Access to data in cloud presents no risk since loading and using the data is allowed only to authenticated users and owners of data, with mapped manner to view the information set together.

# 4. SYSTEM REQUIREMENTS

## 4.1 FUNCTIONAL REQUIREMENTS:

**1) CLIENT**

Here client should register with our application after registration then client should login with the application after successful login view profile, request keys and view keys, new service register for cloud, user cloud services and logout.

**2) ADMIN**

Here admin also should register with the application, here this admin role is assigned by the transaction manager, after that admin can login he can perform some operation such as view profile, view files and can have the change to delete and logout.

**3) SYSTEM MANAGER**

Here manager also should register with the application, here this manager role is assigned by the transaction manager, after that manager can login he can perform some operation such as view profile, upload files and view files and logout.

**4) TRANSACTION MANAGER**

Here manager is a module can directly login with the application after successful login he can perform some operations such as create role and view, create group and view, view all users, view key request, verify and assign group, view service users, view group members and can have the chance to exclude the group member also and logout.

**5) CLOUD**

Here cloud can directly login with the application and after successful login he can perform some operation such as view all uploaded file in the cloud and logout.

## 4.2 NON-FUNCTIONAL REQUIREMENTS

This section elaborates on the functional requirements of the application. The SRS itself can be divided into module, each module having specifications. In order to carry out the project, the following hardware and software is required.

### 4.2.1 HARDWARE REQUIREMENTS

- System : i5 Processor $12^{th}$ Gen
- Hard Disk : 512 Gb
- Ram : 16 Gb

### 4.2.2 SOFTWARE REQUIREMENTS

- Operating System : Windows 11
- Development Software : JAVA 8
- Programming Language : Java
- Integrated Development Environment (IDE) : NetBeans 8.1
- Front End Technologies : HTML5, CSS3, Java Script
- Database Language : SQL
- Database (RDBMS) : MYSQL
- Database Software : MYSQL Server
- Web Server or Development Server : Apache Tomcat
- Design/Modelling : Rational Rose

# 5. SYSTEM DESIGN

## 5.1 FEASIBILITY STUDY

The feasibility of the project is analyzed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

## 5.2 FEASIBILITY ANALYSIS

Three key considerations involved in the feasibility analysis are

- ECONOMICAL FEASIBILITY
- TECHNICAL FEASIBILITY
- SOCIAL FEASIBILITY

## ECONOMICAL FEASIBILITY

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

## TECHNICAL FEASIBILITY

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

## SOCIAL FEASIBILITY

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

## FEASIBILITY ANALYSIS:

### Technical Feasibility:

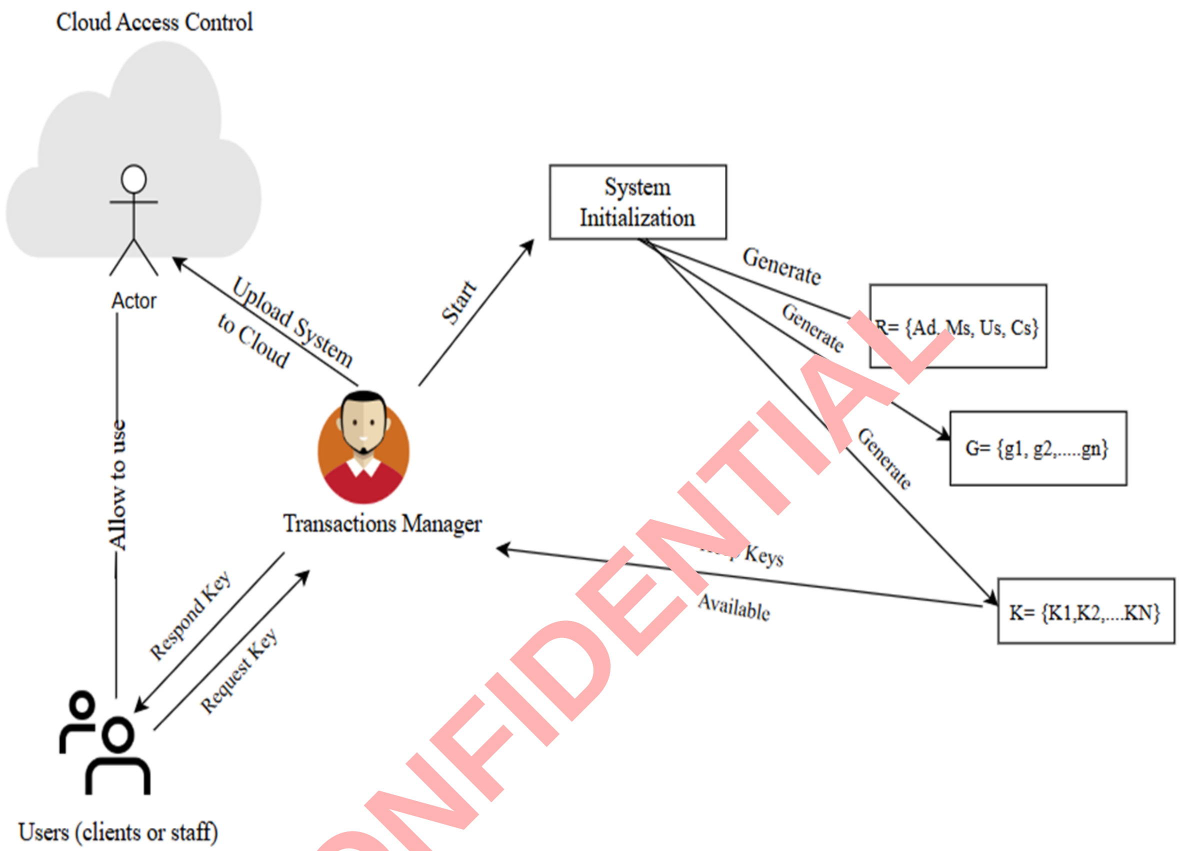
- Technology Stack: The system's technical prerequisites can be readily fulfilled through the utilization of widely accessible technologies, such as data mining libraries, relational databases, and statistical tools. These foundational components collectively form a robust and accessible framework for developing and operating the system.
- Relational databases, exemplified by MySQL, offer scalable and efficient data storage solutions. These databases empower the system to store and manage data in a structured manner, facilitating seamless retrieval and manipulation.
- Incorporating HTML, CSS, and Java into the tech stack is essential for building modern, interactive, and visually appealing web application. Here's why these three technologies are integral to any web development project:
    - **HTML (Hypertext Markup Language):** HTML serves as the structural foundation of web pages. It defines the layout and content structure of your application's user interface. With HTML, you can create various elements like forms, headings, paragraphs, and links. It is essential for presenting content to users in a structured and meaningful way.
    - **CSS (Cascading Style Sheets):** CSS complements HTML by providing styling and design capabilities. It allows you to control the visual presentation of your web application, including aspects such as fonts, colors, layout, and responsiveness. By using CSS, you can ensure that your application is visually appealing, user-friendly, and responsive across different devices and screen sizes.

- **Java (Backend):** Java is a powerful and versatile programming language that is well-suited for building the backend of web applications. It offers strong support for server-side processing, business logic implementation, database connectivity, and handling user requests. Java's scalability, reliability, and robustness make it a popular choice for developing enterprise-grade web applications. Popular Java frameworks like Spring and Java EE provide a structured and efficient way to build the backend of your web application.

- By combining HTML, CSS, and Java, we create a full-stack web development environment that covers both the frontend and backend aspects of the system. This combination allows for seamless communication between the user interface and back-end logic, enabling the creation of dynamic and interactive web application. Moreover, Java's stability and extensive libraries make it a reliable choice for building the core functionality of our system while HTML and CSS ensure an engaging and visually appealing user experience.

# 6. SYSTEM DESIGN

## 6.1 SYSTEM ARCHITECTURE



## 6.2 UML DIAGRAMS

UML stands for Unified Modelling Language. UML is a standardized general-purpose modelling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group. The goal is for UML to become a common language for creating models of object oriented computer software. In its current form UML is comprised of two major components: a Meta- model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.

The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems. The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems. The UML is a very important part of developing objects oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

**GOALS:**

The Primary goals in the design of the UML are as follows:

- Provide users a ready-to-use, expressive visual modelling Language so that they can develop and exchange meaningful models.
- Provide extendibility and specialization mechanisms to extend the core concepts. Be independent of particular programming languages and development process.
- Provide a formal basis for understanding the modelling language.
- Encourage the growth of OO tools market.
- Support higher level development concepts such as collaborations, frameworks, patterns and components.
- Integrate best practices.